

TIME SERIES

Instructor: TRAN THI HA

Forecasting Household Electricity Consumption Using SARIMA

Evidence from Hourly Power Consumption Data

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Abstract

Electricity consumption forecasting is important for energy planning, demand management, and short-term operational decision-making. This project forecasts hourly household electricity consumption using the target variable `Global_active_power`. The analysis uses household power consumption data originally recorded at minute-level frequency and processed into a regular hourly time series. The empirical workflow includes data preprocessing, exploratory data analysis, seasonality analysis, stationarity testing, SARIMA/SARIMAX candidate modelling, baseline comparison, residual diagnostics, and a limited noise robustness check. The final selected model is $\text{SARIMA}(1, 0, 1)(0, 1, 1)_{24}$, which explicitly captures daily seasonal dependence in the hourly series. On the one-month test set, the selected model achieves $\text{MAE} = 0.5948$, $\text{RMSE} = 0.8636$, and $\text{MASE}_{168} = 0.8830$. Since MASE_{168} is below 1, the model improves over the weekly seasonal naive benchmark. However, residual diagnostics are not fully adequate. Residual autocorrelation and heteroskedasticity remain; therefore, the model should be interpreted as a practical forecasting benchmark rather than a fully adequate data-generating model.

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1 Introduction

1.1 Research Context

Electricity consumption forecasting is an important task in energy planning, short-term demand management, and operational decision-making. Electricity demand is not constant over time. It changes systematically according to human activity, appliance usage, working routines, and other recurring behavioural patterns. For household electricity consumption, these patterns are especially visible across different hours of the day and different days of the week.

Hourly household power demand usually contains strong time-dependent structures. Consumption may increase during morning and evening periods, decline during low-activity hours, and vary between weekdays and weekends. These regular patterns make time-series forecasting methods suitable for modelling household electricity demand.

Load forecasting is also important because electricity supply and demand must be balanced continuously. Inaccurate forecasts may increase uncertainty in planning and operation, while effective forecasts can reduce uncertainty and support better energy management decisions [1]. Therefore, a forecasting model for household electricity consumption should be able to capture short-term temporal dependence and recurring seasonal behaviour.

1.2 Research Problem

Household electricity consumption is not random over time. It contains daily and weekly seasonal structures, short-term dependence, and stochastic noise. A forecasting model must capture these temporal patterns while remaining interpretable and empirically justified.

This creates a practical modelling problem. A model that is too simple may fail to capture daily and weekly consumption cycles. A model that is too complex may overfit the training data, become computationally unstable, or produce results that are difficult to interpret. Therefore, this project focuses on SARIMA-type models because they provide a transparent statistical framework for modelling seasonal time-series data.

The main empirical issue is whether SARIMA and SARIMAX specifications can forecast hourly household electricity consumption better than simple naive benchmarks. The analysis also considers whether the selected model remains stable when random noise is introduced into the data.

1.3 Research Question

The main research question of this project is:

How well can SARIMA-type models forecast hourly household electricity consumption after accounting for daily seasonality, weekly structure, and stochastic noise?

This main question is supported by three sub-questions:

- (i) What temporal patterns are present in hourly household electricity consumption?
- (ii) Which SARIMA or SARIMAX specification performs best against naive forecasting benchmarks?
- (iii) How stable is the selected model when random noise is introduced?

1.4 Research Objectives

The objectives of this project are as follows:

- (i) Clean and transform the raw minute-level household power consumption data.
- (ii) Aggregate the observations from minute-level frequency to hourly frequency.
- (iii) Identify trend, seasonality, stationarity, ACF/PACF structure, and noise in the processed time series.
- (iv) Fit and compare baseline, SARIMA, and SARIMAX candidate models.
- (v) Evaluate forecast accuracy on a chronological one-month test set.
- (vi) Diagnose residuals and assess the limitations of the selected model.
- (vii) Conduct a limited noise robustness experiment based on the logic of ARIMA noise-robustness analysis.

1.5 Contribution

This project provides an empirical application of SARIMA-based forecasting to household electricity consumption and evaluates whether the selected model improves upon simple seasonal naive benchmarks. It also includes a limited robustness check inspired by noise-based ARIMA robustness analysis.

The contribution is applied rather than theoretical. The project does not propose a new forecasting model. Instead, it demonstrates a complete empirical workflow for household electricity forecasting, including preprocessing, exploratory analysis, model identification, benchmark comparison, residual diagnostics, and limited robustness testing. This makes the project useful as a transparent forecasting benchmark for structured household electricity consumption data.

1.6 Report Structure

The rest of the report is organized as follows. Section 2 reviews SARIMA forecasting and noise robustness in electrical load forecasting. Section 3 describes the data and preprocessing procedure. Section 4 presents exploratory data analysis and model identification. Section 5 explains the modelling strategy. Section 6 reports the forecasting results. Section 7 discusses diagnostics, robustness, and limitations. Section 8 concludes the report.

2 Literature Review and Theoretical Background

2.1 Electrical Load Forecasting

Electrical load forecasting is a widely studied topic in energy research because electricity consumption varies over time and must be anticipated for planning and operational purposes. Forecasting helps reduce uncertainty in demand management, resource allocation, and short-term energy scheduling. In power systems, load forecasting is especially important because electricity supply and demand need to be balanced continuously [1].

Forecasting tasks are commonly classified according to the forecast horizon. Very short-term load forecasting usually refers to forecasts up to one hour ahead. Short-term load forecasting covers horizons from one hour to several days or weeks. Medium-term forecasting usually concerns longer planning horizons, while long-term forecasting is used for strategic planning. The appropriate forecasting method depends on the objective, the

data frequency, and the forecast horizon.

This project is closest to short-term or medium-short household electricity forecasting. The test horizon contains 720 hourly observations, corresponding to approximately one month. However, the data used in this project describe electricity consumption from a single household, not the aggregate load of a national or regional power system. Therefore, the findings should be interpreted as evidence for household-level consumption forecasting rather than system-level load forecasting.

Household electricity consumption has strong behavioural components. Demand can vary by hour of the day, day of the week, appliance usage, and unobserved household activities. These patterns make household electricity consumption suitable for time-series forecasting methods, but they also introduce noise and irregular variation that may limit model performance.

2.2 ARIMA and SARIMA Models

ARIMA models are among the standard statistical models for time-series forecasting. An ARIMA model combines three components: autoregressive terms, differencing, and moving-average terms. The autoregressive component captures dependence on past values of the series. Differencing is used to remove non-stationarity in the mean. The moving-average component captures dependence on past forecast errors.

The general non-seasonal ARIMA model is denoted as $\text{ARIMA}(p, d, q)$, where p is the autoregressive order, d is the differencing order, and q is the moving-average order. SARIMA extends ARIMA by adding seasonal autoregressive, seasonal differencing, and seasonal moving-average components. The general notation is:

$$\text{SARIMA}(p, d, q)(P, D, Q)_s, \tag{1}$$

where P , D , and Q are the seasonal autoregressive, seasonal differencing, and seasonal moving-average orders, and s is the seasonal period.

For hourly electricity consumption data, daily seasonality is naturally represented by $s = 24$, because each day contains 24 hourly observations. Weekly seasonality can be represented by $s = 168$, because each week contains $24 \times 7 = 168$ hourly observations. In practice, direct modelling of multiple seasonal periods can be computationally demanding. Therefore, one practical approach is to model the dominant daily seasonality using SARIMA and approximate the weekly pattern using additional terms such as Fourier regressors.

Previous research shows that ARIMA-type models are frequently used in electrical load forecasting because they are interpretable and suitable for structured time-series data [1]. Their main advantage is transparency: the role of lagged values, differencing, and seasonal error dependence can be examined directly. Their main limitation is that they are linear models and may not fully capture nonlinear behaviour, sudden regime changes, or unobserved household-level factors.

2.3 Model Identification Using ACF and PACF

ARIMA and SARIMA model identification is usually based on the structure of the time series after suitable transformation and differencing. The autocorrelation function (ACF) and partial autocorrelation function (PACF) are commonly used to identify candidate autoregressive and moving-average orders. This follows the Box–Jenkins modelling logic, in which the researcher studies stationarity, dependence structure, parameter estimation, and residual adequacy before selecting a final model [3].

The ACF measures correlation between the series and its lagged values. The PACF measures the direct relationship between the series and a specific lag after controlling for shorter lags. In the context of hourly electricity consumption, a significant spike at lag 1 suggests short-term dependence. A spike around lag 24 suggests daily seasonal dependence. A spike around lag 168 suggests weekly seasonal dependence.

In this project, ACF and PACF analysis is used together with stationarity tests and observed seasonal patterns. The candidate models are not chosen arbitrarily; they are derived from observed seasonality, stationarity tests, and ACF/PACF patterns. This approach is safer than a purely brute-force search because it connects model selection to the empirical structure of the data.

However, ACF and PACF plots are diagnostic tools rather than mechanical decision rules. Several SARIMA specifications may appear plausible for the same time series. Therefore, candidate models are compared using out-of-sample forecast accuracy, information criteria, convergence behaviour, and residual diagnostics.

2.4 Noise and Robustness

Electricity consumption data contain stochastic variation. At the household level, this variation can come from appliance usage, occupant behaviour, measurement irregularities, holidays, weather conditions, and other unobserved events. Even after preprocessing and seasonal modelling, part of the series may remain noisy and difficult to forecast.

Noise matters because it can affect both parameter estimation and forecast uncertainty. If the series contains substantial random disturbance, the selected ARIMA or SARIMA model may still capture the general seasonal structure, but its parameter estimates and prediction intervals may become less stable. The reference paper by Chodakowska, Nazarko, and Nazarko studies this issue by adding white noise with different noise-to-signal ratios to a reference ARIMA-generated load series, re-estimating the model, and comparing the resulting forecasts [1].

This project applies the same general logic at a smaller empirical scale. After selecting the final SARIMA model, a limited robustness check is conducted by introducing random noise and re-estimating the selected model structure. The purpose is to examine whether forecast accuracy and prediction interval width remain stable when additional disturbance is added to the training series.

The robustness analysis in this project is limited because only the low-noise case was successfully estimated; higher-noise cases encountered memory constraints. Therefore, the robustness results should be interpreted cautiously. They provide preliminary evidence on low-level noise sensitivity, not a complete robustness analysis across all disturbance levels.

2.5 Literature Gap and Project Positioning

Existing studies show that SARIMA-type models are suitable for structured electricity load data, especially when the series contains clear seasonal patterns. The literature also shows that preprocessing, model identification, and noise sensitivity are important issues in ARIMA-based load forecasting.

This project does not aim to propose a new forecasting model. Instead, it applies a transparent SARIMA modelling workflow to household electricity consumption and evaluates its practical forecasting performance, residual adequacy, and limited noise sensitivity. The project is positioned as an empirical forecasting exercise using household-level hourly data. Its contribution is the careful application and evaluation of SARIMA-type models against naive benchmarks, rather than the development of a new forecasting method.

3 Data and Preprocessing

3.1 Data Source

This project uses the *Individual Household Electric Power Consumption* dataset, which records electricity consumption from a single household located in Sceaux, France. The original dataset contains minute-level measurements of household power consumption, voltage, current intensity, and three sub-metered appliance groups. Since the objective of this project is to forecast short-term household electricity consumption, the target variable is `Global_active_power`, measured in kilowatts.

Although the original public dataset covers a longer period, the working dataset used in this project contains 1,048,575 minute-level observations from 16 December 2006 17:24:00 to 13 December 2008 21:38:00. This working sample is sufficient for SARIMA modelling because it contains nearly two years of continuous high-frequency observations, allowing the model to capture both daily and weekly seasonal patterns.

3.2 Variable Description

The dataset includes seven numerical measurement variables after removing the original date and time columns. The variables are described in Table 1.

Table 1: Variable description

Variable	Description	Unit	Role in this project
<code>Global_active_power</code>	Household global active power consumption	kW	Main forecasting target
<code>Global_reactive_power</code>	Household global reactive power	kW	Descriptive variable
<code>Voltage</code>	Minute-averaged voltage	V	Descriptive variable
<code>Global_intensity</code>	Household global current intensity	A	Descriptive variable
<code>Sub_metering_1</code>	Energy sub-metering for kitchen appliances, including dishwasher, oven, and microwave	Wh	Descriptive variable
<code>Sub_metering_2</code>	Energy sub-metering for laundry-room appliances, including washing machine, tumble dryer, refrigerator, and light	Wh	Descriptive variable
<code>Sub_metering_3</code>	Energy sub-metering for water heater and air conditioner	Wh	Descriptive variable

The final forecasting model is built mainly as a univariate time-series model using `Global_active_power`. The remaining variables are retained for understanding the structure of household electricity consumption, but they are not used as direct regressors in the final SARIMA model.

3.3 Initial Data Inspection

The raw data initially contained 1,048,575 rows and 9 columns, including `Date`, `Time`, and seven measurement variables. Most numerical columns were imported as object or string variables because missing values were represented by the character `?`. Therefore, direct numerical modelling was not possible before cleaning.

The initial inspection showed three main data issues. First, the `Date` and `Time` columns were stored separately and had to be combined into a single timestamp. Second, missing

measurement values were represented both as ? and as NaN. Third, the numerical columns had to be converted from object type to floating-point format before aggregation and modelling.

There were 4,069 rows with missing measurement values, equal to approximately 0.39% of the working sample. No duplicated timestamps were found after constructing the datetime index.

3.4 Datetime Construction

The `Date` and `Time` columns were merged into a single `Datetime` variable using the format `dd/mm/yyyy hh:mm:ss`. This new datetime variable was then set as the index of the dataframe. The original `Date` and `Time` columns were removed because their information was already contained in the datetime index.

This step is necessary for time-series forecasting because SARIMA models require observations to be ordered by time. A proper datetime index also allows resampling from minute-level frequency to hourly frequency.

3.5 Missing Value Treatment

Missing values represented by ? were first replaced with NaN. After this replacement, all measurement variables were converted to numeric floating-point format. Linear interpolation was then applied to fill missing observations.

Linear interpolation is appropriate in this setting because the dataset is a continuous high-frequency time series and the share of missing observations is small. It preserves the temporal continuity required for SARIMA modelling. This is important because ARIMA-type models are sensitive to gaps and irregular spacing in the time index. Previous research on ARIMA-based load forecasting also emphasizes that the preprocessing stage is important because poorly treated measurement data and random disturbances can affect model identification and predictive performance.

After interpolation, all seven measurement variables contained zero missing values.

3.6 Hourly Resampling

The original data were recorded at one-minute frequency. For this project, the data were resampled to hourly frequency using the mean value within each hour. The processed

hourly dataset contains 17,477 observations, covering the period from 16 December 2006 17:00:00 to 13 December 2008 21:00:00.

Hourly aggregation is used for three reasons. First, hourly electricity forecasting is more aligned with short-term load forecasting applications than minute-level forecasting. Second, aggregation reduces short-term random fluctuations and computational burden. Third, the hourly frequency allows clear modelling of daily seasonality with a 24-hour period and weekly seasonality with a 168-hour period.

For the target variable, `Global_active_power`, the hourly value is interpreted as the average active power consumption within each hour. This is consistent with the objective of forecasting average hourly household power demand.

3.7 Final Processed Dataset

After preprocessing, the final hourly dataset has 17,477 rows and 7 numerical columns. No missing values remain in the processed data. The target variable, `Global_active_power`, has a mean of 1.1057 kW, a standard deviation of 0.9671 kW, a minimum of 0.1240 kW, and a maximum of 6.5605 kW.

Table 2: Summary statistics of the processed hourly target variable

Statistic	<code>Global_active_power</code>
Count	17,477
Mean	1.1057
Standard deviation	0.9671
Minimum	0.1240
25th percentile	0.3101
Median	0.7672
75th percentile	1.6200
Maximum	6.5605

These statistics show that household electricity consumption is right-skewed: most hourly observations are relatively low, while some hours contain high-consumption peaks. This pattern is expected in household electricity data because appliance usage is irregular and depends on household behaviour.

3.8 Train-Test Split

The processed time series was split chronologically into training and testing sets. Random splitting was not used because it would leak future information into the training process and produce unreliable forecasting results.

The last 720 hourly observations were reserved as the test set. This corresponds to approximately one month of hourly forecasts. The remaining observations were used for model training.

Table 3: Train-test split

Set	Period	Number of observations
Training set	2006-12-16 17:00:00 to 2008-11-13 21:00:00	16,757
Test set	2008-11-13 22:00:00 to 2008-12-13 21:00:00	720
Total	2006-12-16 17:00:00 to 2008-12-13 21:00:00	17,477

This chronological split allows the model to be estimated only on past data and evaluated on unseen future observations, which is consistent with the forecasting objective of the project.

3.9 Preprocessing Summary

The preprocessing procedure can be summarized as follows. The raw minute-level data were loaded and inspected. The **Date** and **Time** columns were merged into a single datetime index. Missing values coded as ? were replaced with **NaN**, and all measurement columns were converted to numeric format. Linear interpolation was applied to handle missing values. The data were then resampled from minute-level to hourly frequency using the mean value within each hour. Finally, the processed hourly dataset was split chronologically into training and testing sets.

This preprocessing pipeline produces a clean, regular, hourly time series suitable for SARIMA-based forecasting. It preserves the temporal order of observations, removes missing-value problems, reduces excessive short-term noise, and prepares the target variable at a frequency where daily and weekly seasonal patterns can be meaningfully analysed.

4 Exploratory Data Analysis and Model Identification

This section summarizes the main empirical patterns in the processed hourly household electricity consumption series and explains how they support the SARIMA/SARIMAX modelling strategy.

4.1 Distribution Analysis

The target variable is `Global_active_power`, measured in kilowatts. Table 4 reports the summary statistics of the hourly target series.

Table 4: Summary statistics of hourly `Global_active_power`

Statistic	Value
Count	17,477
Mean	1.1057
Standard deviation	0.9671
Minimum	0.1240
Median	0.7672
Maximum	6.5605
Skewness	1.3224
Kurtosis	1.6882

The distribution is right-skewed. Most hourly observations are relatively low, while a smaller number of observations correspond to high-consumption periods. Since the dataset does not contain behavioural information, these peaks are interpreted descriptively rather than causally.

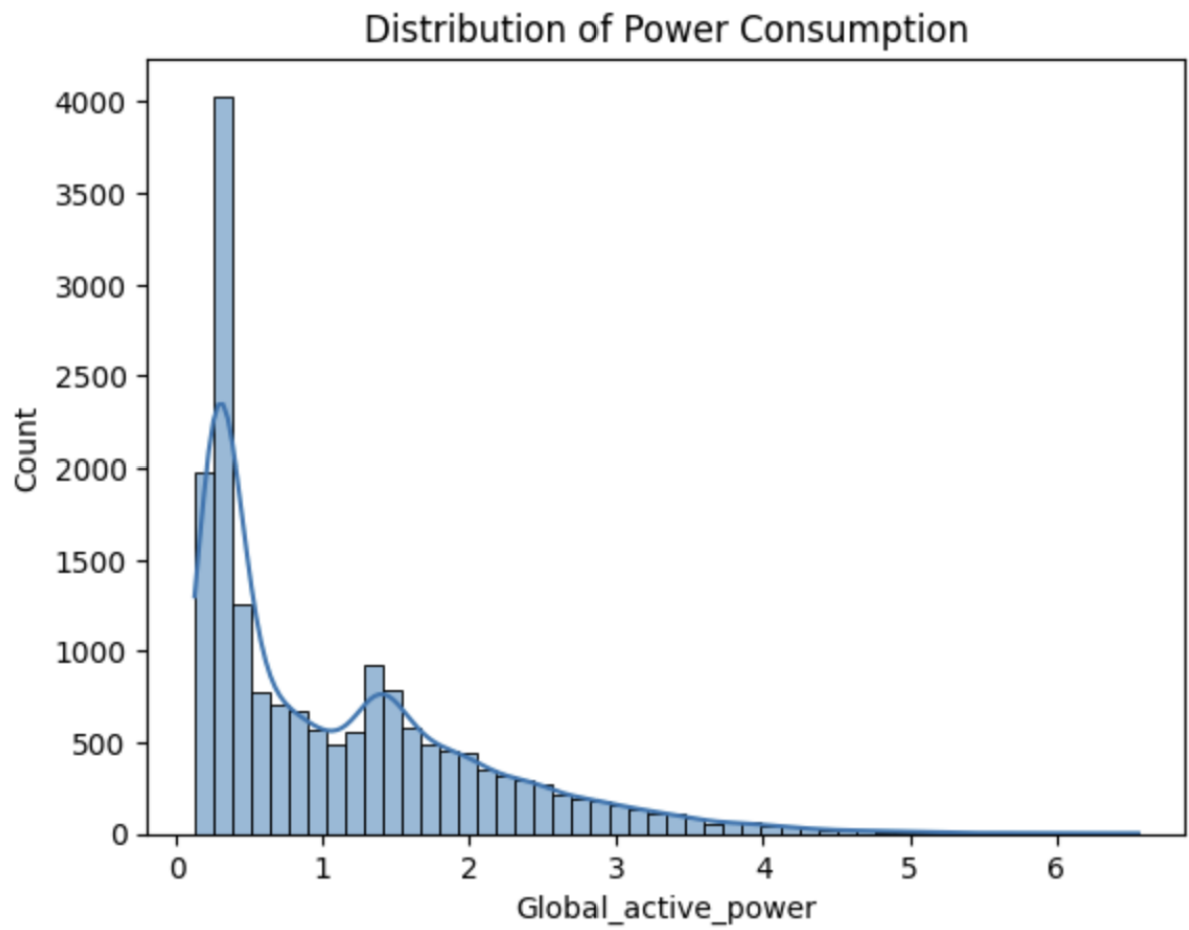


Figure 1: Distribution of hourly `Global_active_power`. The series is right-skewed, with a concentration of low-to-moderate observations and a smaller number of high-consumption peaks.

4.2 Time-Series Structure and Seasonality

The full hourly time-series plot shows repeated temporal structure rather than a strong monotonic trend. A zoomed weekly view shows clear daily cycles, while rolling statistics indicate that both the local mean and variance change over time. This suggests that recurring seasonal behaviour is more important than long-term trend.

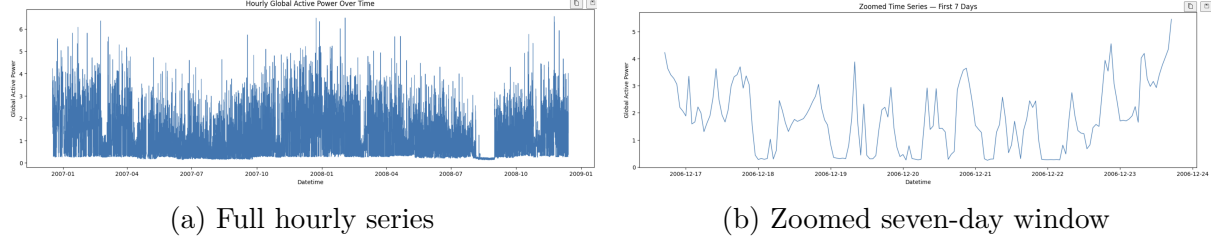


Figure 2: Hourly household electricity consumption over time. The full series shows repeated temporal structure, while the weekly window makes the daily cycle more visible.

Seasonality is also visible across hours of the day and days of the week. Hour-of-day patterns confirm intraday variation, while day-of-week patterns indicate weekly behavioural differences. A day-hour heatmap further supports repeated daily consumption patterns. Therefore, the model should account for a 24-hour daily cycle and consider a 168-hour weekly cycle.

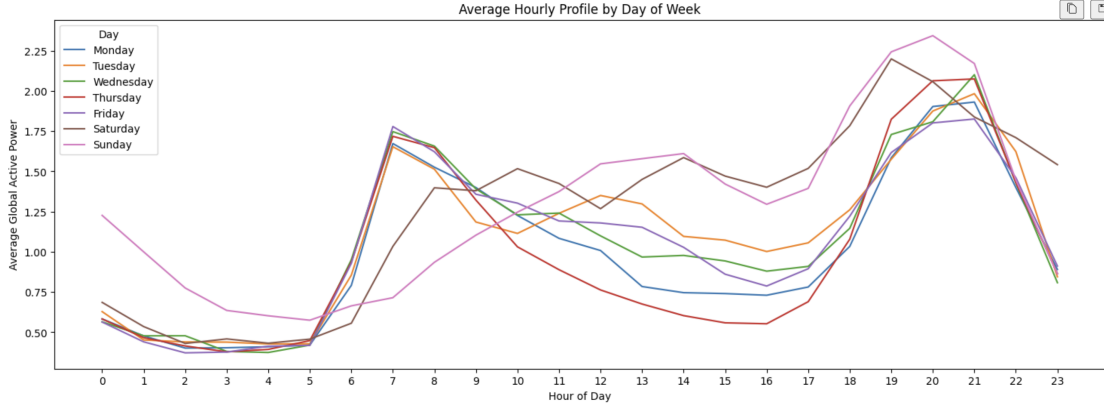


Figure 3: Average hourly consumption profiles. Intraday differences support a 24-hour seasonal component, while variation across days of the week supports the presence of weekly structure.

4.3 Periodogram and Phase-Space Evidence

The periodogram identifies strong repeating periods around 24 hours and 168 hours. The main seasonal peaks include approximately 24.01 hours and 168.05 hours, supporting daily and weekly seasonality. Additional peaks around 12 hours suggest within-day sub-cycles.

Phase-space plots at lags 1, 24, and 168 show dependence between current consumption and lagged consumption. Lag 1 indicates short-term dependence, lag 24 indicates daily repetition, and lag 168 indicates weekly repetition. These patterns support SARIMA-type modelling because the series has structured seasonal dependence rather than independent random fluctuations.

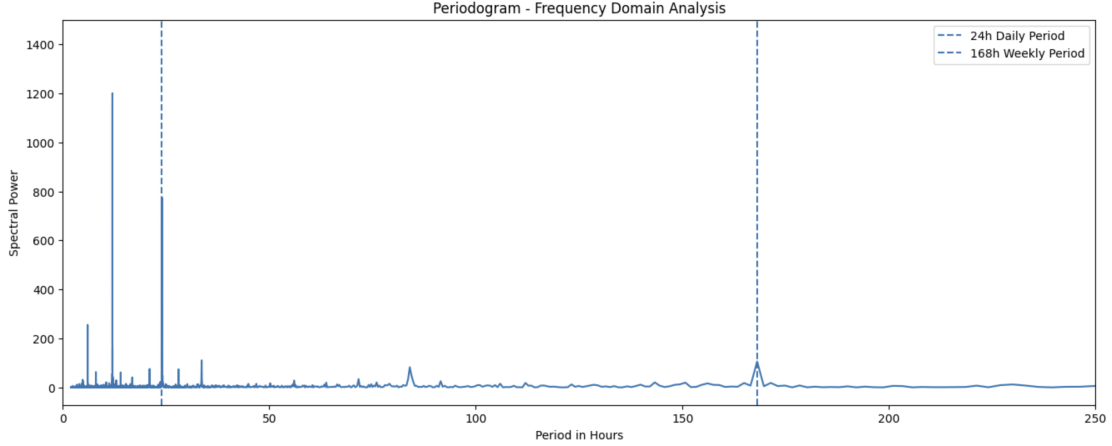


Figure 4: Periodogram of hourly `Global_active_power`. The marked peaks around 24 hours and 168 hours indicate daily and weekly seasonal structure.

4.4 Stationarity Tests

Stationarity is assessed using both the Augmented Dickey–Fuller (ADF) test and the KPSS test. The ADF test for the original series rejects the unit-root null hypothesis, with statistic -11.4476 and p-value close to zero. However, the KPSS test rejects stationarity, with statistic 1.9408 and p-value 0.01 . This mixed result is common in seasonal electricity data: the series may not contain a strong unit root, but it still contains clear seasonal structure and changing local behaviour.

Since strong daily and weekly seasonality is present, differencing remains justified for SARIMA identification. The modelling notebook tests first differencing, seasonal differencing at lag 24, seasonal differencing at lag 168, and combinations of these transformations. For practical SARIMA implementation, the modelling stage uses $d = 1$, $D = 1$, and $s = 24$, while weekly structure is considered through benchmark comparison and Fourier-type approximations. The final differencing decision is based on both statistical tests and the observed seasonal structure, not on a single test result.

4.5 ACF/PACF Interpretation

The original ACF/PACF plots show clear temporal dependence and seasonality. After differencing, the ACF/PACF plots indicate shorter-memory dependence. Behaviour around lag 1 supports testing non-seasonal AR/MA terms, while behaviour around lag 24 supports including a daily seasonal component, especially a seasonal moving-average term. Dependence around lag 168 confirms weekly structure, but this is handled indirectly due to computational constraints.

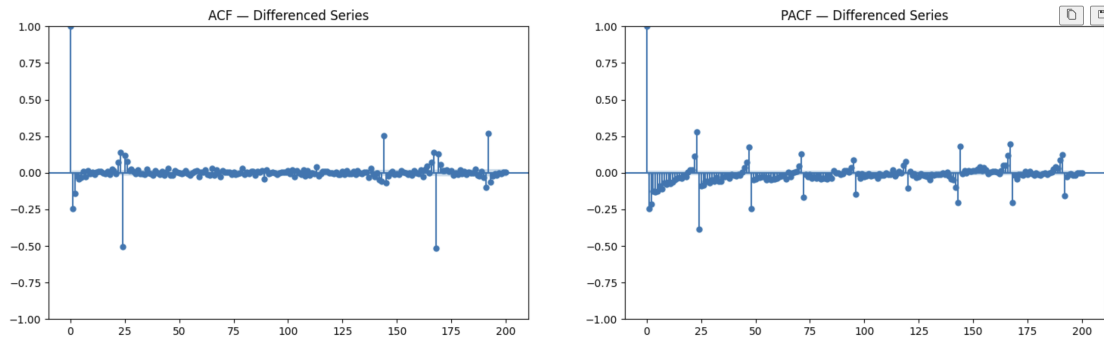


Figure 5: ACF and PACF of the differenced hourly series. Short-lag dependence and seasonal behaviour around lag 24 motivate the tested SARIMA specifications.

The candidate SARIMA/SARIMAX models are therefore derived from the observed dependence structure rather than selected through arbitrary brute-force search.

4.6 Noise Analysis

Noise is examined using STL decomposition with a 24-hour seasonal period. The decomposition separates the series into trend, seasonal, and residual components. The residual component is approximately centered around zero, but its variance changes over time, indicating remaining stochastic variation and possible heteroskedasticity.

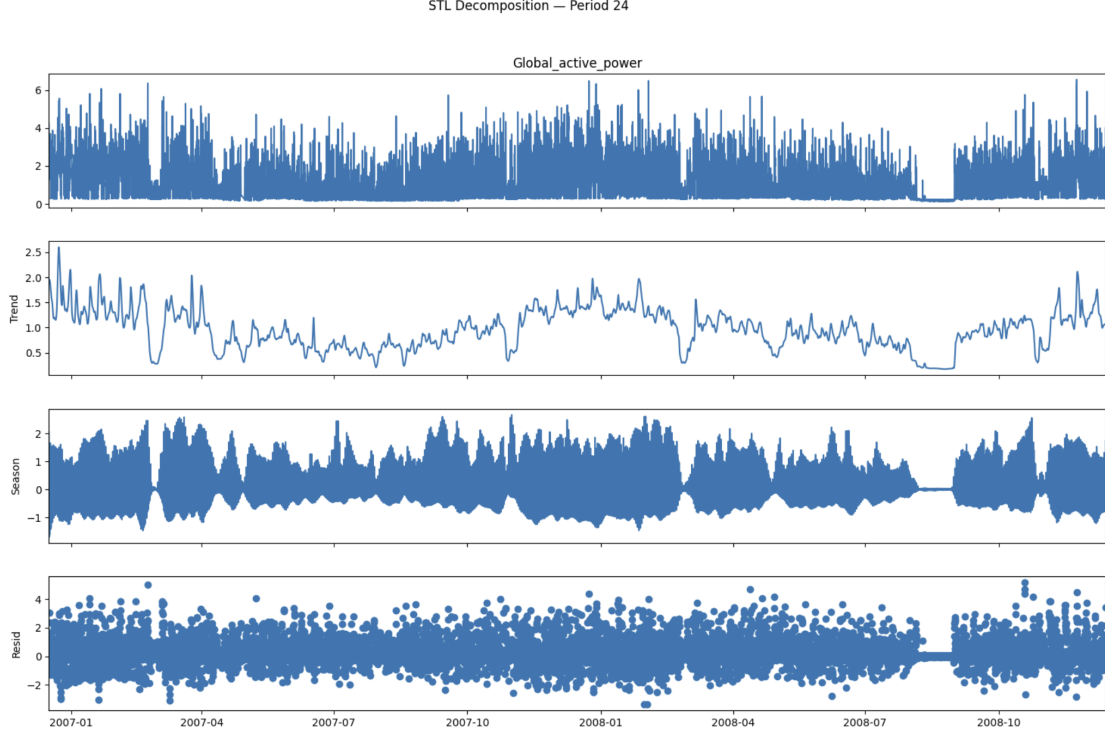


Figure 6: STL decomposition of hourly `Global_active_power` using a 24-hour seasonal period. The residual component shows remaining stochastic variation after extracting the main daily pattern.

The noise-to-signal ratio is calculated as:

$$\text{NSR} = \frac{\sigma_{\text{noise}}}{\sigma_{\text{signal}}}. \quad (2)$$

Using the STL residual standard deviation as σ_{noise} , the estimated NSR is approximately 0.7149, or 71.49%. This indicates a moderate level of stochastic variation relative to the total variability of the target series. The estimated NSR is used as an exploratory indicator, not as a definitive measurement of measurement error.

4.7 EDA-Based Model Identification Summary

The EDA shows that hourly household electricity consumption is dominated by recurring daily and weekly structures rather than a strong monotonic trend. The ACF/PACF plots indicate short-term and seasonal dependence, while STL decomposition shows that residual stochastic variation remains after extracting the daily pattern.

These findings justify SARIMA-type modelling. The practical modelling stage focuses on SARIMA specifications with daily seasonality ($s = 24$), while weekly structure is

considered through the weekly seasonal naive benchmark and selected SARIMAX/Fourier variants.

5 Methodology

This section describes the forecasting framework used in the project. The methodology follows a structured empirical workflow: benchmark models are estimated first, SARIMA/SARIMAX candidate models are estimated second, and the final model is selected based on out-of-sample forecasting accuracy, convergence behaviour, and residual diagnostics. Information criteria such as AIC and BIC are reported, but they are not used as the only selection criterion because the main objective is forecasting performance on unseen data.

5.1 Modelling Framework

The forecasting task is defined as a univariate time-series problem using hourly `Global_active_power` as the target variable. Let y_t denote household active power consumption at hour t . The objective is to forecast future values of y_t over the chronological test period.

The modelling framework consists of four stages. First, simple benchmark models are estimated to provide reference forecasts. Second, a set of theoretically motivated SARIMA and SARIMAX candidate models is fitted on the training set. Third, the candidate models are evaluated on the one-month test set using forecast accuracy metrics. Fourth, the selected model is examined using residual diagnostics and a limited noise robustness experiment.

The final model is selected according to the following criteria:

- (i) out-of-sample forecast accuracy on the chronological test set;
- (ii) improvement over simple seasonal naive benchmarks;
- (iii) successful convergence during estimation;
- (iv) reasonable information criteria, including AIC and BIC;
- (v) residual diagnostic results, including autocorrelation and heteroskedasticity checks.

AIC and BIC are useful for comparing in-sample model fit and parsimony, but they are not sufficient by themselves. A model with a low AIC or BIC may still perform poorly

on unseen observations or produce inadequate residual diagnostics. Therefore, the final model is selected using both statistical and forecasting criteria.

5.2 Baseline Models

Baseline models are included to provide simple reference points for evaluating SARIMA performance. A SARIMA model is useful only if it improves upon simple seasonal naive forecasts.

Three baseline models are used in this project: a naive last-value forecast, a daily seasonal naive forecast, and a weekly seasonal naive forecast. Their purposes are summarized in Table 5.

Table 5: Baseline forecasting models

Model	Purpose
Naive last value	Minimal benchmark that assumes future consumption remains equal to the latest observed value
Seasonal naive 24h	Daily repetition benchmark that uses the value from the same hour on the previous day
Seasonal naive 168h	Weekly repetition benchmark that uses the value from the same hour in the previous week

The naive last-value model is defined as:

$$\hat{y}_{T+h} = y_T, \quad (3)$$

where T is the final observation in the training set and h is the forecast horizon.

The daily seasonal naive forecast is defined as:

$$\hat{y}_t = y_{t-24}, \quad (4)$$

where the forecast for a given hour is based on the same hour of the previous day.

The weekly seasonal naive forecast is defined as:

$$\hat{y}_t = y_{t-168}, \quad (5)$$

where the forecast for a given hour is based on the same hour of the previous week. This benchmark is especially important because the exploratory analysis shows evidence of weekly structure in household electricity consumption.

5.3 Candidate SARIMA and SARIMAX Models

A small set of theoretically motivated SARIMA/SARIMAX candidates was estimated, including non-seasonal AR/MA terms, daily seasonal components with $s = 24$, and selected weekly Fourier regressors.

The general SARIMA model is denoted as:

$$\text{SARIMA}(p, d, q)(P, D, Q)_s, \quad (6)$$

where p , d , and q are the non-seasonal autoregressive, differencing, and moving-average orders; P , D , and Q are the seasonal autoregressive, seasonal differencing, and seasonal moving-average orders; and s is the seasonal period.

In operator form, the SARIMA model can be written as:

$$\Phi_P(B^s)\phi_p(B)(1 - B)^d(1 - B^s)^D y_t = \Theta_Q(B^s)\theta_q(B)\varepsilon_t, \quad (7)$$

where B is the backshift operator, $\phi_p(B)$ and $\theta_q(B)$ are non-seasonal autoregressive and moving-average polynomials, $\Phi_P(B^s)$ and $\Theta_Q(B^s)$ are seasonal autoregressive and moving-average polynomials, and ε_t is the error term.

The candidate model groups include the following specifications:

Table 6: Candidate SARIMA/SARIMAX model groups

Model group	Purpose
SARIMA(0, 1, 1)(0, 1, 1) ₂₄	Baseline daily seasonal moving-average structure
SARIMA(1, 1, 1)(0, 1, 1) ₂₄	Adds short-term autoregressive dependence
SARIMA(1, 0, 1)(0, 1, 1) ₂₄	Tests daily seasonal structure without non-seasonal differencing
SARIMA(2, 1, 1)(0, 1, 1) ₂₄	Tests a richer short-term autoregressive structure
SARIMA(1, 1, 2)(0, 1, 1) ₂₄	Tests a richer short-term moving-average structure
SARIMAX variants with Fourier terms	Approximates weekly seasonality using exogenous Fourier regressors

The candidate models are derived from the exploratory analysis rather than selected arbitrarily. The ACF/PACF results indicate short-term dependence around lag 1 and seasonal dependence around lag 24. The periodogram and phase-space plots also indicate weekly structure around lag 168. Therefore, the modelling strategy focuses on daily SARIMA components and selected weekly approximations.

5.4 Weekly Fourier Terms

Standard SARIMAX models can efficiently handle one main seasonal period. In this project, daily seasonality is represented directly using the SARIMA seasonal period $s = 24$. Weekly seasonality is more difficult to include directly because a true multiple-seasonal SARIMA model with both $s = 24$ and $s = 168$ can become computationally expensive.

As a practical compromise, weekly seasonality is approximated using Fourier terms with period 168. For a weekly period $m = 168$, the Fourier regressors are defined as:

$$\sin\left(\frac{2\pi kt}{168}\right) \quad \text{and} \quad \cos\left(\frac{2\pi kt}{168}\right), \quad (8)$$

where k is the Fourier order and t is the time index.

These Fourier terms allow the SARIMAX model to capture smooth weekly cyclical patterns without estimating a full second seasonal ARIMA component. Weekly Fourier terms approximate the 168-hour cycle; they do not fully replace a true double-seasonal SARIMA model. Therefore, results from SARIMAX models with Fourier terms should be interpreted as a practical approximation rather than a complete multiple-seasonal specification.

5.5 Evaluation Metrics

Forecast performance is evaluated using several complementary metrics. Let y_t be the observed value, $\hat{y}_t$ be the forecast value, and n be the number of observations in the test set.

Mean absolute error (MAE) measures average absolute forecast error:

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|. \quad (9)$$

Root mean squared error (RMSE) penalizes larger errors more heavily:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2}. \quad (10)$$

Symmetric mean absolute percentage error (sMAPE) provides a percentage-type relative error measure:

$$\text{sMAPE} = \frac{100}{n} \sum_{t=1}^n \frac{|y_t - \hat{y}_t|}{(|y_t| + |\hat{y}_t|)/2}. \quad (11)$$

Weighted absolute percentage error (WAPE) measures total absolute error relative to total observed consumption:

$$\text{WAPE} = 100 \times \frac{\sum_{t=1}^n |y_t - \hat{y}_t|}{\sum_{t=1}^n |y_t|}. \quad (12)$$

MASE with weekly seasonality compares the model against a weekly seasonal naive benchmark. With $m = 168$, it is defined as:

$$\text{MASE}_{168} = \frac{\frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|}{\frac{1}{T-m} \sum_{t=m+1}^T |y_t - y_{t-m}|}, \quad (13)$$

where T is the number of training observations and $m = 168$ is the weekly seasonal period.

A MASE_{168} value below 1 indicates improvement over the weekly seasonal naive benchmark. This metric is important because the weekly seasonal naive model is a strong and interpretable baseline for hourly household electricity consumption.

AIC and BIC are also reported for SARIMA/SARIMAX models. They measure in-sample fit adjusted for model complexity. Their general forms are:

$$\text{AIC} = -2\ell + 2k, \quad (14)$$

$$\text{BIC} = -2\ell + k \ln(n), \quad (15)$$

where ℓ is the maximized log-likelihood and k is the number of estimated parameters.

The metrics used in this project are summarized in Table 7.

Table 7: Evaluation metrics

Metric	Purpose
MAE	Measures average absolute forecast error
RMSE	Penalizes larger forecast errors more heavily
sMAPE	Measures relative percentage-type forecast error
WAPE	Measures aggregate absolute error relative to total observed consumption
MASE_{168}	Compares performance against the weekly seasonal naive benchmark
AIC/BIC	Measures in-sample model fit and parsimony

5.6 Residual Diagnostics

Residual diagnostics are used to assess whether the selected model has adequately captured the structure of the time series. Forecast accuracy alone is not sufficient, because a model can perform reasonably on the test set while still leaving systematic structure in the residuals.

The residual diagnostics include:

- (i) residual time plot;
- (ii) residual ACF/PACF plots;
- (iii) Ljung–Box test for residual autocorrelation;
- (iv) ARCH test for conditional heteroskedasticity;
- (v) prediction interval width analysis.

The residual time plot is used to inspect whether the residuals are centered around zero and whether their variance changes over time. The residual ACF/PACF plots show whether autocorrelation remains after fitting the model.

The Ljung–Box test checks whether residual autocorrelation remains jointly significant across a set of lags. The null hypothesis is that the residuals are independently distributed up to the tested lag. A small p-value indicates that residual autocorrelation remains.

The ARCH test checks whether the variance of the residuals changes systematically over time. The null hypothesis is no ARCH effect. A small p-value indicates conditional heteroskedasticity.

Prediction interval width is also examined because forecast uncertainty is an important part of time-series forecasting. The average 95% prediction interval width is calculated as:

$$\text{Mean PI Width} = \frac{1}{n} \sum_{t=1}^n (U_t - L_t), \quad (16)$$

where U_t and L_t are the upper and lower bounds of the prediction interval.

5.7 Noise Robustness Experiment

A limited noise robustness experiment is conducted to examine whether the selected SARIMA model remains stable when random disturbance is introduced into the training series. The experiment follows the general logic of ARIMA noise-robustness analysis in the reference literature [1].

Gaussian white noise is added to the training series as follows:

$$\tilde{y}_t = y_t + \eta_t, \quad (17)$$

where $\tilde{y}_t$ is the noise-disturbed series and η_t is Gaussian white noise:

$$\eta_t \sim N(0, \sigma_{\text{noise}}^2). \quad (18)$$

The noise standard deviation is defined as:

$$\sigma_{\text{noise}} = \text{NSR} \times \sigma_{\text{signal}}, \quad (19)$$

where NSR is the noise-to-signal ratio and σ_{signal} is the standard deviation of the original training signal.

After the noise-disturbed training series is created, the selected SARIMA structure is re-estimated. Forecast errors and prediction interval width are then compared with the base model. The main comparison focuses on MAE, RMSE, sMAPE, WAPE, MASE_{168} , and mean prediction interval width.

The experiment follows the logic of the reference paper but is limited by memory constraints in the current implementation. As a result, the robustness analysis should be interpreted as a limited sensitivity check rather than a complete simulation study across all possible noise levels.

6 Results

This section presents the empirical results in four steps. First, the baseline forecasting models are evaluated. Second, the SARIMA/SARIMAX candidate models are compared. Third, the final selected model is interpreted. Fourth, residual diagnostics and the limited noise robustness experiment are reported. The results are linked directly to the main research question: how well SARIMA-type models forecast hourly household electricity consumption after accounting for seasonal structure and stochastic variation.

6.1 Baseline Results

Table 8 reports the forecasting performance of the three benchmark models on the one-month chronological test set.

Table 8: Baseline forecasting results

Model	MAE	RMSE	sMAPE	WAPE	MASE_168
Naive last value	0.8138	1.0219	60.44%	56.36%	1.2081
Seasonal naive 24h	0.7609	1.1053	62.53%	52.69%	1.1295
Seasonal naive 168h	0.7071	0.9678	51.69%	48.97%	1.0497

Among the simple benchmarks, the weekly seasonal naive model performs best. It has the lowest MAE, sMAPE, WAPE, and MASE_168 among the baseline models. This confirms the importance of weekly repetition in the hourly household electricity consumption series. However, its MASE_168 value remains above 1, meaning that there is still room for improvement over the simple weekly repetition benchmark.

6.2 SARIMA/SARIMAX Model Comparison

Table 9 compares the estimated SARIMA/SARIMAX candidates. The table reports out-of-sample forecast errors, information criteria, convergence status, and boundary validity. The boundary check is included because models with parameters too close to non-stationary or non-invertible boundaries may be unstable, even if their forecast error appears competitive.

Table 9: SARIMA/SARIMAX candidate model comparison

Model	MAE	RMSE	sMAPE	WAPE	MASE_168	AIC	BIC	Boundary OK
SARIMA_B_(1,1,1)(0,1,1,24)	0.6146	0.8257	48.81%	42.56%	0.9123	30385.47	30416.37	No
SARIMA_E_(2,1,1)(0,1,1,24)	0.6149	0.8261	48.85%	42.58%	0.9128	30374.21	30412.84	No
SARIMA_C_(1,0,1)(0,1,1,24)	0.5948	0.8636	45.38%	41.19%	0.8830	30470.75	30501.65	Yes
SARIMAX_I_(1,1,1)(0,0,1,24)+F168K3	0.7904	1.0216	60.67%	54.74%	1.1733	32674.49	32751.75	No
SARIMA_D_(1,1,1)(0,0,1,24)	0.7948	1.0380	60.93%	55.04%	1.1798	32715.27	32746.17	No
SARIMA_A_(0,1,1)(0,1,1,24)	1.1309	1.3914	129.58%	78.32%	1.6787	32912.65	32935.83	Yes

Some models achieve competitive in-sample information criteria or low RMSE but are not preferred because they have boundary problems or weaker overall out-of-sample performance. In particular, **SARIMA_B** and **SARIMA_E** produce the lowest RMSE values, but their maximum absolute AR/MA parameters are close to the boundary, so they are treated as less stable candidates.

Several candidate models with Fourier regressors or richer short-memory structures could not be fully estimated because of memory allocation limits. These failed estimations are

reported as computational limitations rather than empirical model failures.

Table 10: Candidate models not completed due to memory constraints

Model	Status
SARIMA_F_(1,1,2)(0,1,1,24)	Failed due to memory allocation limit
SARIMAX_G_(1,1,1)(0,1,1,24)+F168K3	Failed due to memory allocation limit
SARIMAX_H_(1,0,1)(0,1,1,24)+F168K3	Failed due to memory allocation limit
SARIMAX_J_(1,1,1)(1,1,1,24)+F168K3	Failed due to memory allocation limit
SARIMAX_K_(1,1,1)(0,1,1,24)+F168K5	Failed due to memory allocation limit

The model comparison shows that selection cannot be based only on AIC or BIC. Forecast accuracy, convergence, parameter stability, and residual behaviour must be considered together.

6.3 Final Model Selection

The final selected model is:

$$\text{SARIMA}(1, 0, 1)(0, 1, 1)_{24}. \quad (20)$$

In the modelling notebook, this model is recorded as:

$$\text{SARIMA_C_}(1, 0, 1)(0, 1, 1, 24).$$

Table 11 reports the forecasting performance of the selected model.

Table 11: Final selected model performance

Metric	Value
MAE	0.5948
RMSE	0.8636
sMAPE	45.38%
WAPE	41.19%
MASE_168	0.8830
AIC	30470.75
BIC	30501.65

The selected SARIMA model improves over the weekly seasonal naive benchmark because its MASE_168 value is below 1. It also has lower MAE, sMAPE, and WAPE than all three baseline models. This supports the use of SARIMA-type modelling for the hourly household electricity consumption series.

The selected model is not chosen because it has the lowest RMSE among all fitted models. It is selected because it provides the best valid balance between forecast accuracy, boundary stability, and interpretability. This is safer than selecting a model that gives a slightly lower RMSE but has unstable parameter behaviour.

6.4 Parameter Interpretation

Table 12 reports the main estimated parameters of the selected SARIMA model.

Table 12: Estimated parameters of the selected SARIMA model

Parameter	Coefficient	p-value	Interpretation
AR(1)	0.6386	< 0.001	Significant short-term dependence
MA(1)	0.0169	0.085	Weak effect, not significant at the 5% level
Seasonal MA(24)	-0.9445	< 0.001	Strong daily seasonal effect
σ^2	0.3604	< 0.001	Residual variance

The AR(1) coefficient is positive and statistically significant, indicating that current hourly consumption is related to recent consumption. The seasonal MA(24) coefficient is

large in magnitude and highly significant, confirming the importance of the daily cycle. However, the MA(1) term is weak and not statistically significant at the 5% level. This suggests that not all included short-term components are equally important.

6.5 Forecast Plot and Interval Analysis

The final model forecasts the one-month test set and produces 95% prediction intervals. Figure 7 should show the actual test observations, the SARIMA forecasts, and the corresponding 95% prediction interval.

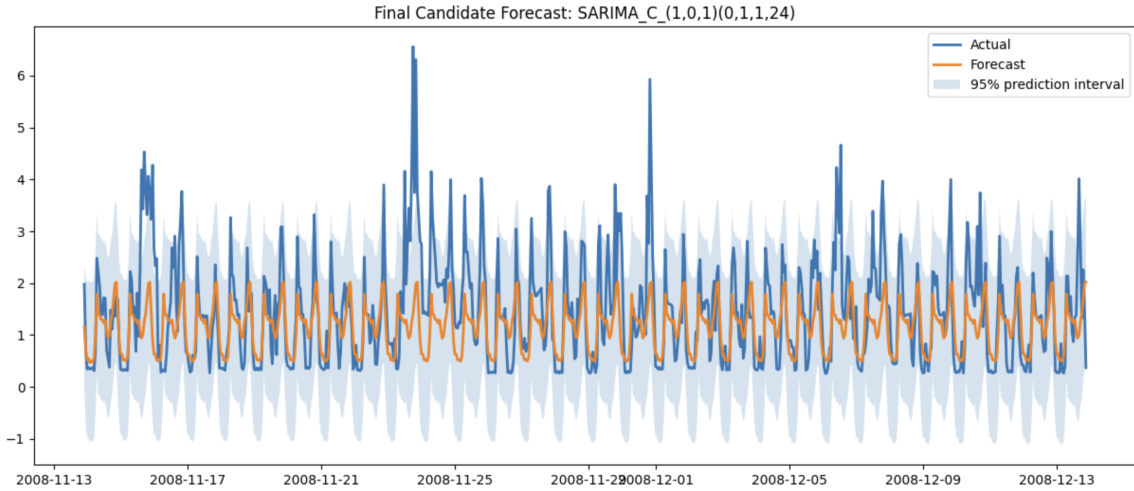


Figure 7: Actual and forecasted hourly household electricity consumption on the one-month test set. The shaded area represents the 95% prediction interval.

The mean 95% prediction interval width is:

$$\text{Mean 95\% PI Width} = 3.1576. \quad (21)$$

The prediction intervals are relatively wide compared with the scale of hourly household consumption. This reflects remaining uncertainty and residual variation. The result is consistent with the exploratory noise analysis, which indicated that the series contains substantial stochastic variation even after seasonal structure is considered.

6.6 Residual Diagnostics

Residual diagnostics are used to evaluate whether the final SARIMA model fully captures the dependence structure of the data. Table 13 reports the Ljung–Box test results at selected lags.

Table 13: Ljung–Box test results for residual autocorrelation

Lag	Test statistic	p-value
1	8.6521	3.2669×10^{-3}
24	373.2851	2.2303×10^{-64}
48	461.8691	5.0060×10^{-69}
168	966.0767	2.9863×10^{-112}

The Ljung–Box test does not pass at the tested lags. The p-values at lags 24, 48, and 168 are extremely small, indicating that residual autocorrelation remains after fitting the model.

Table 14 reports the ARCH test results for conditional heteroskedasticity.

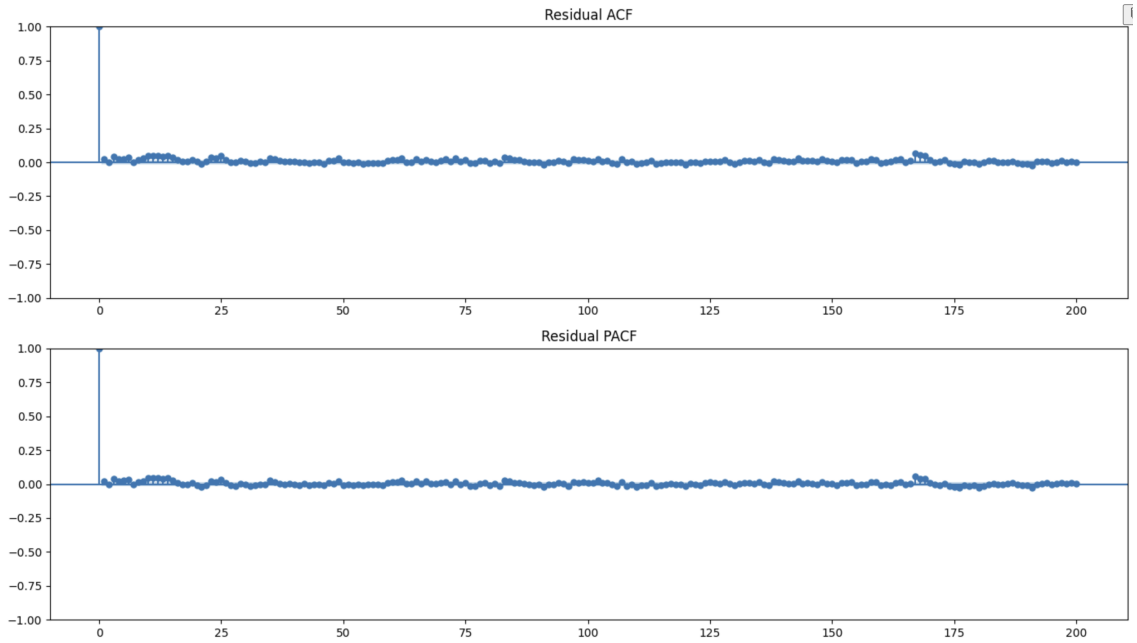


Figure 8: Residual diagnostics for the selected SARIMA model. Remaining autocorrelation and non-constant residual variance are consistent with the Ljung–Box and ARCH test results.

Table 14: ARCH test results

Test	Value
ARCH LM statistic	8.3420×10^2
ARCH LM p-value	9.1401×10^{-173}
ARCH F statistic	8.7735×10^1
ARCH F p-value	2.5999×10^{-177}

The ARCH test also does not pass. The extremely small p-values indicate remaining heteroskedasticity in the residuals.

Residual diagnostics are not fully adequate. The model leaves residual autocorrelation and heteroskedasticity. Therefore, the final SARIMA model should be reported as a forecasting benchmark, not as a complete statistical representation of the data-generating process.

6.7 Noise Robustness Result

The robustness experiment adds Gaussian white noise to the training series and re-estimates the selected SARIMA structure. Table 15 compares the base model with the noise-disturbed cases.

Table 15: Noise robustness results

NSR	σ_{noise}	MAE	RMSE	sMAPE	MASE_168	Mean CI Width	Status
0.0	0.0000	0.5948	0.8636	45.38%	0.8830	3.1576	Base
0.1	0.0903	0.5972	0.8632	45.58%	0.8865	3.1765	Completed
1.0	0.9026	NaN	NaN	NaN	NaN	NaN	Failed due to mem
2.0	1.8051	NaN	NaN	NaN	NaN	NaN	Failed due to mem

At $\text{NSR} = 0.1$, the model remains close to the base model. MAE increases only slightly from 0.5948 to 0.5972, while RMSE remains almost unchanged. The mean confidence interval width increases from 3.1576 to 3.1765, suggesting only a small increase in forecast uncertainty under low noise.

However, the robustness experiment is incomplete because the higher NSR cases failed due to memory allocation limits. Therefore, no strong conclusion should be made about

robustness at high noise levels. The results only support a limited conclusion: the selected SARIMA structure appears stable under the low-noise case tested here, but robustness under stronger random disturbance remains unresolved.

6.8 Results Summary

The empirical results answer the research question in a qualified way. SARIMA-type modelling improves over simple naive benchmarks, especially the weekly seasonal naive benchmark. The selected model, $\text{SARIMA}(1, 0, 1)(0, 1, 1)_{24}$, achieves the best valid overall forecasting performance among the tested candidates and produces a MASE_{168} below 1.

However, the residual diagnostics show that the model does not fully remove autocorrelation or heteroskedasticity. The robustness analysis is also limited by computational constraints. Therefore, the final model should be interpreted as a practical forecasting benchmark for hourly household electricity consumption, not as a fully adequate statistical model of the underlying consumption process.

7 Discussion

This section interprets the main findings, relates them to the literature, and states the main limitations of the selected model.

7.1 Interpretation of Findings

The results show that household electricity consumption has clear daily and weekly temporal structure. The EDA identifies repeated patterns around the 24-hour and 168-hour periods, while the model comparison shows that a SARIMA model with daily seasonality improves over simple naive benchmarks.

The selected model, $\text{SARIMA}(1, 0, 1)(0, 1, 1)_{24}$, achieves $\text{MASE}_{168} = 0.8830$, which is below 1. This means that it improves over the weekly seasonal naive benchmark. Therefore, SARIMA-type modelling is useful for this hourly household electricity consumption series when daily seasonality is explicitly represented.

These results are consistent with previous research showing that ARIMA/SARIMA models are useful statistical tools for electricity load forecasting because they can model temporal dependence, differencing, and seasonal behaviour [1, 3]. The daily and weekly

patterns are also consistent with household electricity routines, appliance usage, and repeated behavioural schedules.

The selected model is practically useful because it is transparent and interpretable. It provides a stronger benchmark than simple repetition-based forecasts and can be used as a reference point for more advanced forecasting methods. However, its practical significance should not be overstated. It should not be treated as a highly accurate real-time deployment model or as a complete explanation of household electricity demand.

7.2 Limitations

This project has several limitations. First, the dataset covers one household only, so the findings may not generalize to households with different behaviours, appliances, or occupancy patterns. Second, the working sample covers December 2006 to December 2008, not the full original dataset. Third, weather, holidays, household occupancy, and appliance-level behavioural variables are not included as exogenous predictors.

Fourth, a true double-seasonal SARIMA model with both 24-hour and 168-hour seasonal periods was not fully implemented due to computational limits. Daily seasonality is modelled directly, while weekly structure is handled through benchmark comparison and selected approximations.

Fifth, the residual diagnostics are not fully satisfactory. The selected SARIMA model fails the Ljung–Box tests, indicating remaining residual autocorrelation. It also fails the ARCH test, indicating conditional heteroskedasticity and time-varying residual variance. For this reason, the model is explicitly positioned as a *practical forecasting benchmark*, not as an ideal *data-generating model*. It can evaluate whether a transparent seasonal statistical model improves over simple benchmarks, but it should not be treated as a complete explanation of household electricity consumption.

Sixth, the noise robustness experiment is incomplete. The low-noise case was successfully estimated, but higher NSR cases failed due to memory constraints. Therefore, the project cannot make strong claims about robustness under high noise levels. Seventh, the forecast horizon is one month only, so the conclusions may differ for shorter or longer horizons.

7.3 Future Work

Future work should address the limitations above. True multiple-seasonal models, TBATS, or MSTL-based models could represent daily and weekly seasonality more directly. Exoge-

nous variables such as temperature, holidays, calendar indicators, and occupancy proxies should be added to explain variation that past consumption alone cannot capture.

The selected SARIMA model should also be compared with ETS, Prophet, XGBoost, LSTM, and hybrid models using rolling-origin cross-validation. Since the model fails the Ljung–Box and ARCH diagnostics, future models should explicitly handle remaining autocorrelation and changing variance. Suitable extensions include ARIMAX–GARCH models or nonlinear multi-seasonal frameworks such as MSTL combined with LSTM networks.

8 Conclusion

This project investigates whether SARIMA-type models can forecast hourly household electricity consumption after accounting for daily seasonality, weekly structure, and stochastic variation. The results show that SARIMA-type models can forecast hourly household electricity consumption better than simple seasonal naive benchmarks when daily seasonality is explicitly modelled.

The main empirical result is that the selected SARIMA(1, 0, 1)(0, 1, 1)₂₄ model achieves MAE = 0.5948, RMSE = 0.8636, and MASE₁₆₈ = 0.8830 on the one-month test set. Since MASE₁₆₈ is below 1, the model improves over the weekly seasonal naive benchmark. This supports the use of SARIMA as a practical statistical forecasting method for structured household electricity consumption data.

However, the residual diagnostics show remaining autocorrelation and heteroskedasticity. Therefore, the selected model is not a complete statistical explanation of the household electricity consumption process. It captures important seasonal structure, especially the daily cycle, but it does not fully explain all systematic variation in the data.

Overall, the project supports SARIMA as a transparent and useful forecasting benchmark for structured household electricity consumption. Further modelling is needed for stronger statistical adequacy, better treatment of weekly seasonality, inclusion of external predictors, and more reliable robustness analysis under higher noise levels.

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